

Typhoon, rainfall and trees jointly cause landslides in coastal regions

Yu Zhuang^a, Aiguo Xing^{a,*}, Yuehua Jiang^b, Qiang Sun^b, Jinkai Yan^c, Yanbo Zhang^a

^a State Key Laboratory of Ocean Engineering, Shanghai Jiao Tong University, Shanghai 200240, PR China

^b Nanjing Center, China Geological Survey, Nanjing 210016, PR China

^c Chinese Academy of Geological Science, Beijing 100037, PR China

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ABSTRACT

Typhoon-induced landslides are the primary natural hazard in hilly coastal regions of South East Asia and can potentially devastate coastal communities and infrastructure. They occur at approximately the same time as typhoon strikes, showing complex failure mechanisms under the coupled action of wind, rainfall and trees. Therefore, special assessments should be performed for such geohazards. Here we elucidated the general features of typhoon-induced landslides in southeast coastal China in terms of their geological characteristics, spatial and temporal distribution, and vegetation cover characteristics. We further established a framework to investigate their possible failure mechanisms coupled to wind, rainfall and trees. Specifically, the mechanical response of trees in a typhoon was estimated through a combination of the autoregressive method (AR) and a multi-degree-of-freedom tree swaying model, while the trees and rainfall contributions were determined by calculating the cohesion attributable to roots and the soil shear strength reduction caused by the water content variation. Our results suggest that the root reinforcement disappears when the sliding surface is below the root embedded depth, and the impact of wind load is significant during the slope stability analysis. The Force 14 wind (mean wind velocity up to 45 m/s) reduces the factor of safety (F_s) of the slope by 20% when blowing downhill, and its combined contribution with rainwater infiltration will lead to a higher risk of slope failures (F_s reduction of 45% at the base of the root-soil composite). Additionally, typhoon-induced shallow slope failures are prone to develop at the base of the root-soil composite or the rock-soil interface. Our model results suggest that the impacts of wind and vegetation on the generation of landslides during a typhoon can be significant and deserve more attention in hazard assessment studies.

1. Introduction

Global climate change induced by mankind activities has substantially increased the number of extreme weather events (Bellprat et al., 2019; Sippel et al., 2020). In recent years, once-in-a-century or in-decades typhoons have occurred frequently, generated numerous landslides, and caused mass casualties and economic losses worldwide (Choy et al., 2020; Sajjad et al., 2020). Typhoons form primarily in the marine environment and propagate from the coastal region to the inland with a constantly decreasing power. Thus, most geohazards are generated in the hilly coastal region during the typhoon under the coupled action of strong winds and heavy rainfall (Chou et al., 2017). Interestingly, typhoon-induced landslides mainly develop in the trees-covered slope, showing differences of disaster-causing factors and failure mechanisms with rainfall-induced landslides (Yu et al., 2006; Wu et al., 2011; Vassallo et al., 2015; Wu et al., 2021; Zhuang et al., 2021). The prevention

and mitigation of disasters caused by such geohazards is a novel and urgent problem.

Many efforts have been made on the triggering mechanism of typhoon-induced landslides through numerical methods, model tests, and field investigations (e.g. Ouyang et al., 2018; Zhang et al., 2019; Shou and Lin, 2020; Tsou et al., 2021). Nevertheless, these efforts mainly focused on case studies and only addressed the impact of rainstorms on slope stability. These studies regarded typhoon-induced landslides as rainfall-induced landslides and ignored the effects of wind and vegetation. Notably, powerful typhoon loads can be transmitted to the slope by acting on the trees and affecting slope stability (Buma and Johnson, 2015; Mohsen and Behzad, 2019). Trees themselves must also be considered during landslide risk assessments because of their mechanical (surcharge load and root reinforcement) and hydrological effects (soil moisture reduction) on slope stability.

Vegetation has been widely used to control landslides worldwide

* Corresponding author.

E-mail address: xingaiguo@sjtu.edu.cn (A. Xing).

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owing to root reinforcement and hydrological effects. The presence of roots results in an increase in soil cohesion through the mechanical resistance of roots in tension and friction with soil (Wu, 1984; Nguyen et al., 2019; Masi et al., 2021). This phenomenon allows for reinforced soils to have a significantly larger shear strength than soils without roots. In terms of hydrological effects, the tree canopy can create an “umbrella effect” to attenuate the rainfall permeate into the soil (Fathizadeh et al., 2017; Gonzalez-Ollauri and Mickovski, 2017), while plant water transpiration is well known to reduce soil saturation and potentially improve soil shear strength (Ni et al., 2017). However, there is a substantial divergence in the extent to which vegetation reduces landslides. The wind load (typhoon) and weight of vegetation can greatly reduce slope stability, especially for high trees with large frontal areas (Li et al., 2021). These two forces increase the driving force acting on the potential sliding surface and decrease the slope stability. Furthermore, previous studies have shown the limitations of vegetation stabilization behavior. The root reinforcement effect is only available within the root zone, and the beneficial hydrological effects (rainfall interception and transpiration) potentially disappear following heavy rain, implying that vegetation can hardly improve the slope stability below the root zone during a rainstorm (Simon and Collison, 2002; McGuire et al., 2016). Due to the complex coupling relationship between vegetation and slopes, the role of vegetation in slope stability is likely variable under different conditions. Thus far, no targeted research with general implications considering the impact of typhoons and vegetation has been conducted on typhoon-induced landslides.

This study aims to establish a general framework for the failure mechanism analysis of typhoon-induced landslides and provide broader

implications for their risk assessment. We first investigated 303 typhoon-induced landslides in the coastal region of southeastern China and summarized their general features combined with existing published data. A novel generalized model of typhoon-induced landslides was then established to further investigate their failure mechanisms coupled with the typhoon, rainfall and trees. The new quantitative insights into the impacts of typhoons on slope stability open new perspectives for the study of typhoon-induced landslides and will eventually assist in landslide risk assessment in hilly coastal regions worldwide.

2. Typhoon-induced landslides

Among the provinces in southeast coastal China, Zhejiang ($27^{\circ}02' - 31^{\circ}11'N$, $118^{\circ}01' - 123^{\circ}10'E$) and Fujian ($23^{\circ}52' - 28^{\circ}37'N$, $115^{\circ}85' - 120^{\circ}67'E$) are the two provinces seriously affected by typhoons (Fig. 1a). Both regions have a subtropical monsoon climate and generate many geohazards in hilly zones during the typhoon season. Landslides are the most frequent and severe cascading effects in the typhoon-damaged region and have caused mass casualties and economic losses in recent decades. China Geological Survey investigated 303 typhoon-induced landslides in Zhejiang coastal regions, and we summarized here their general features according to investigation results combined with the existing published data (investigation employed by Yan et al., 2020 in Fujian).

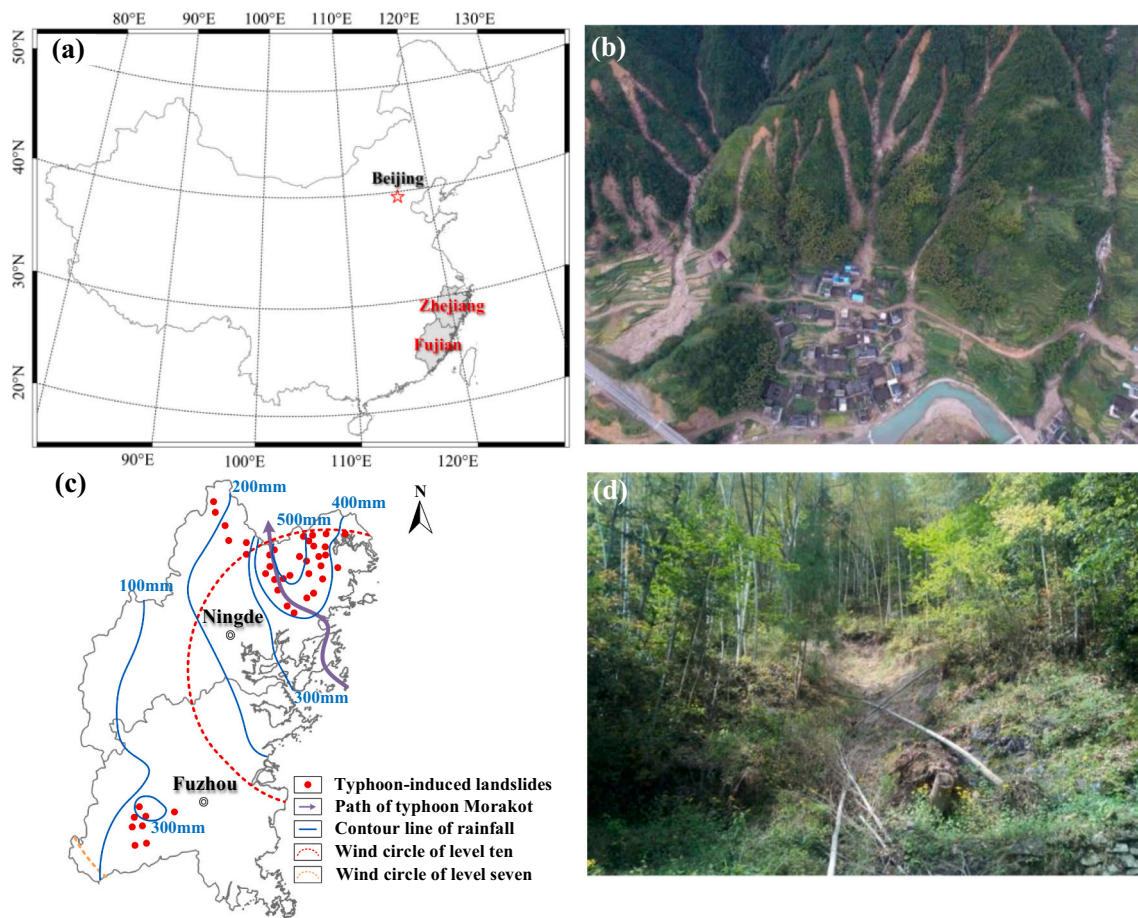


Fig. 1. a Locations of Zhejiang and Fujian provinces. b A cluster of typhoon-induced landslides. c Distribution of typhoon Morakot-induced landslides in the northern Fujian. Wind circles of level seven and ten imply the region with a mean wind velocity reaches 13.9–17.1 m/s (Force 7 wind) and 24.5–28.4 m/s (Force 10 wind), respectively. d An example of the typhoon-induced landslide generated in the trees-covered slope, showing the heavily forested hillslope.

2.1. Geomorphological and geological characteristics of typhoon-induced landslides

Among the 303 investigated landslides, 68% of the events were shallow landslides with a depth <3 m and a volume <1000 m³, and only a few landslides had a maximum depth >10 m and a total volume >10,000 m³. In addition, 70% of typhoon-induced landslides had a slope angle of 20°–40°, and the frequency reached a peak value of approximately 30°. In terms of the rock mass structure, the slopes are characterized by a two-layer structure, implying that weathering-formed loose deposits overlay the bedrock. Thus, sliding surfaces primarily develop at the interface between the bedrock and overburden or a possible weak layer in the overburden soil mass. Notably, the sliding mass mainly consists of Quaternary loose deposits with the composition of silt, clay and silty clay that contains gravel and had poor cohesion.

2.2. Spatial distribution of typhoon-induced landslides

The spatial distribution of typhoon-induced landslides is highly consistent with the typhoon path and its affected area. The wind caused by typhoons could reach more than the Beaufort scale force (hereinafter named Force) 12 at the typhoon center when it propagates to the coastal region. Most secondary landslides were clustered in trees-covered slopes (Fig. 1b) within the radii of seven and ten-level wind circles, indicating the critical role of typhoons (Yan et al., 2020). Here, wind circles of level seven and ten imply the region with a mean wind velocity reaches 13.9–17.1 m/s (Force 7 wind) and 24.5–28.4 m/s (Force 10 wind), respectively (Geng et al., 2018). We also confirmed the consistency between landslide frequency and rainfall intensity, as clusters of landslides occurred in regions with high rainfall intensity. For instance, landslides induced by Typhoon Morakot in 2009 presented a prevalence in zones within the ten-level wind circle and a cumulative rainfall of more than 200 mm (Fig. 1c). Furthermore, few landslides have been identified in the coastal plain areas because of their flat topography. Typhoon-induced landslides are prone to develop in the hilly coastal regions because of frequent human activities (e.g. slope excavation) and steep terrain for the landslide occurrence.

2.3. Temporal distribution of typhoon-induced landslides

Typhoon-induced landslides have a weak time-lag effect and happen about the same time as typhoon strikes. Our investigation results indicated that 87% of the landslides failed within 12 h after typhoon landing and were in the duration of the windstorm and heavy rainfall. This is also the peak period of the wind force and rainfall intensity, implying the special temporal features of typhoon-induced landslides.

2.4. Vegetation characteristics of typhoon-induced landslides

All the investigated events occurred on slopes with high vegetation coverage (Fig. 1d). Various vegetation types were identified in the landslide masses. Most were trees (pine and fir), followed by bamboo and shrubs. During the typhoon process, powerful wind loads can be transmitted to the slope by acting on trees with large frontal areas and affecting slope stability as the driving force.

3. Methods

Typhoon-induced landslides show a unique failure mechanism because of the coupled action of wind, rainfall, and trees. Thus, their contributions (both stabilizing and destabilizing the slope) should be taken into account when conducting the slope stability analysis. This study established a slope stability model that accounts for the wind load, hydrological effects of rainfall (through soil strength reduction), and root contribution to analyze the possible failure mechanism of typhoon-induced landslides.

3.1. Typhoon impact

The wind force transferred to the slope was determined by calculating the mechanical response of trees in the wind field, which can act as a driving force to lower slope stability. Numerical wind field modeling is integrated as a static and fluctuating component (Elshare et al., 2017). The Power-law model, which suggested that the wind speed variation depended on the surface roughness coefficient α , was used to estimate the mean wind profile (Mayhoub and Azzam, 1997):

$$\frac{v_z}{v_{10}} = \left(\frac{z}{z_{10}} \right)^\alpha \quad (1)$$

where z represents the height from the ground, v_{10} is the mean wind speed measured at 10 m height (z_{10}), and could be determined based on the measurement or Beaufort scale (Elizabeth and Peter, 1997), α is related to the environment, and a value of 0.25–0.3 is recommended for a trees-covered slope (Gualtieri and Secchi, 2014; Hadi, 2015).

The fluctuating wind velocity can be regarded as a zero-mean stochastic process that satisfies a specific wind power spectrum. Here, the process $u(t)$ is calculated using the classical autoregressive (AR) model (Santamaría-Bonfila et al., 2016), described as follows:

$$u(t) = \sum_{j=1}^p \psi_j u(t - j\Delta t) + Ln(t) \quad (2)$$

where ψ_j are autoregressive parameters, L is a lower triangular matrix, $n(t)$ is made of normally distributed random numbers with zero mean and unit variance, and p is the order of the model that reflects the calculation accuracy ($p=4$ can satisfy most simulations).

Multiplying $u^T(t - j\Delta t)$ on both sides of Eq. (2), calculating the expectation and obtaining the matrix of the autoregressive parameters:

$$R(j\Delta t) = \sum_{j=1}^p \psi_j R((j-k)\Delta t) \quad (3)$$

$$LL^T = R(0) - \sum_{j=1}^p \psi_j R(k\Delta t) \quad (4)$$

$R(j\Delta t)$ is the autocorrelation function of the fluctuating wind and satisfies the Wiener-Khinchine theorem:

$$R(j\Delta t) = \int S(z, n) r_{ij} \cos(2\pi n j \Delta t) dn \quad (5)$$

where r_{ij} is the cross-spectral density function that denotes the correlations between two points (Paola, 1998), and $S(n)$ is the empirical spectrum model. Notably, the general Kaimal spectrum model (Kaimal et al., 1972) that varies with height is selected because of the slender shape of trees. The combination of Eqs. (3)–(5) determine the autoregressive parameters (ψ_j) and the random term ($Ln(t)$), and then the wind speed time series can be obtained according to Eq. (2). More details regarding the AR modeling process are presented in Yu et al. (2017).

We modeled the mechanical response of trees in a typhoon using a modified multi-degree-of-freedom tree swaying model (Pivato et al., 2014). In this study, we summarized the main features of the original tree-motion model. Trees are represented as flexible cantilever beams with equal diameters that are completely clamped in the ground. A linear modal analysis was used for the tree motion calculation by composing the tree deformation into a set of vibration modes so that the tree displacement was the sum of the contributions of each vibration mode. Sellier et al. (2008) pointed out that the first vibration mode, which corresponds to the first vibration frequency, is the main vibration mode, as its contribution is far ahead of the other vibration modes. Thus, only the first vibration mode was considered during the calculation in this study. The wind force on the tree accounted for large tree deflections by considering the geometric nonlinearities because of the

inclination of the flexible beam relative to the wind direction. The tree-to-tree crown collision was neglected because the contact forces between trees were negligible compared with the wind force (Py et al., 2006; Pivato et al., 2014). Particularly, the contribution of eccentric tree gravity is significant because of the large deformation of trees under powerful typhoons, and a model with real geometric features will also improve the modeling accuracy (Neild and Wood, 1999; Yang et al., 2005). The tree has a crown, and the diameter of the trunk and crown decreases with height, causing the beam with a constant diameter to be too simplified for a tree model (Fig. 2). Therefore, we modified the tree model as a cantilever beam with variable cross-sections closer to the geometric features of the trees, and added the term eccentric gravity (Fig. 2c) to the dynamic equation. Moreover, the inertia force caused by tree motion was considered when analyzing the load transmitted to the soil. A detailed schematic representation of the modified tree model is shown in Fig. 2. Fig. 2a shows a real tree to show readers its geometric features, while Fig. 2b is a simplified cantilever model established based on Fig. 2a and used for the numerical work in this study. The trunk and crown diameters decrease linearly with height so that the gravity and frontal area of each segment also decrease with height. Subsequently, the gravity and wind force acting on each tree segment can be calculated based on the assumed diameter and frontal area distribution, and a modified multi-degree-of-freedom tree swaying model can be established (Fig. 2c). The dynamic equations of tree motion and the expression of the wind force are as follows:

$$m \frac{\partial^2 y}{\partial t^2} + c \frac{\partial y}{\partial t} + ky = \int_0^{h_t} F_{wi} \phi ds + \int_0^{h_t} G_i \phi ds \quad (6)$$

$$F_{wi} = \rho C_d A_f \left| u \cos \theta_i - \frac{\partial y}{\partial t} \cos \theta_i \right| \left(u \cos \theta_i - \frac{\partial y}{\partial t} \cos \theta_i \right) \cos \theta_i \quad (7)$$

$G_i = m_i g \cdot \sin \theta_i \cdot \cos \theta_i$ (8), where m , c , k , ϕ are the first modal mass, damping, stiffness, and mode shape, respectively, which are determined using a finite-element method, y is its associated generalized displacement, F_{wi} is the wind force and G_i is the contribution of the eccentric beam gravity of each segment. h_t is the tree height, ρ is the air density, C_d is the drag coefficient (Pivato et al., 2014), A_f is the frontal area, u is the wind velocity, and θ_i is the inclination of the segment relative to the vertical direction.

Pines universally distributed in southeast coastal China were selected for the modeling and calculation, while the corresponding input parameters were generalized based on investigations by Wang (2014), as presented in Table 1. Fig. 2d shows the calculated first mode shape and associated vibration frequency. The mechanical response of the trees in a typhoon was deduced by introducing the wind velocity calculated using Eqs. (6) and (7), and subsequently using the central finite-difference scheme to solve the equations. The total force-time series at the tree trunk base under a Force 10 typhoon is presented in Fig. 2e (shown as an example). The positive and negative values of the wind force are directional values. We chose this combined model in this study because it does not require a large number of unconstrained parameters,

Table 1

Model parameters used for the numerical simulations of the tree response and slope stability.

Definition	Unit	Value
Surface roughness coefficient α	–	0.3
Order of AR model p	–	4
Tree trunk mass	kg	350
Tree crown mass	kg	55
Tree height h_t	m	20
Crown width	m	5
Crown height	m	10
Diameter at the base of the tree trunk	m	0.25
Tree distance L	m	3.5
Root depth	m	2
First modal mass m	kg	11.2
First modal damping c	kg/s	10.6
First modal stiffness k	kg/s ²	62.8
Vibration frequency of the first mode	Hz	0.372
Drag coefficient C_d	–	0.35
Length of beam segment ds	m	1
Friction angle ϕ	°	16–25
Soil cohesion (in the natural state) c_{sn}	kPa	9–15
Slope angle β	°	30
Depth of root-soil composite h_{root}	m	2
Overburden depth	m	3
Soil unit weight γ	kN/m ³	19
Natural volumetric water content of soil ω_n	–	0.38
Saturated volumetric water content of soil ω_s	–	0.52
Strength reduction coefficient μ	–	6.94

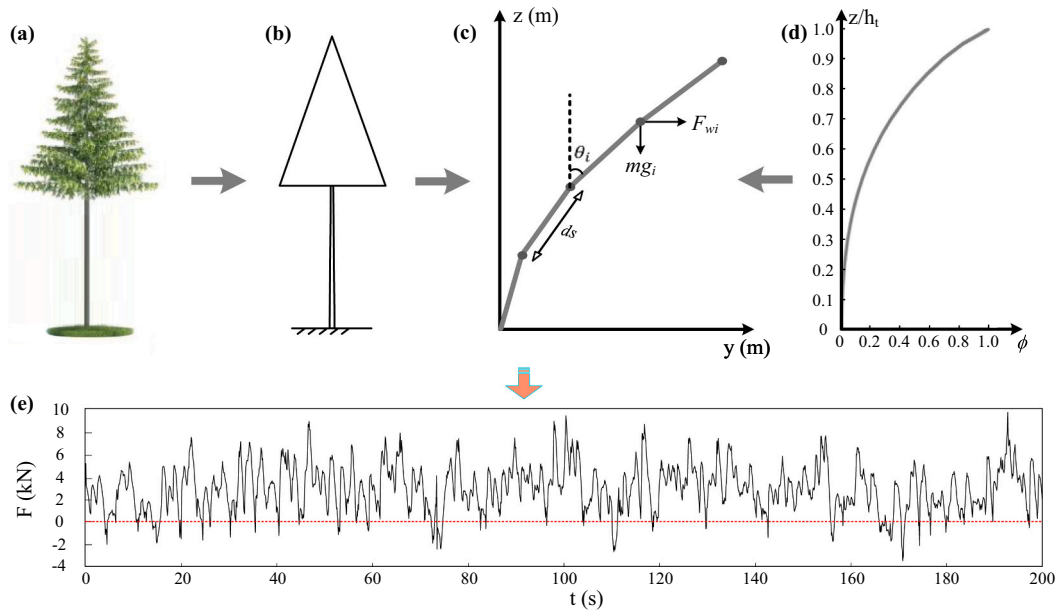


Fig. 2. a–c Modeling tree as a multi-degree-of-freedom cantilever beam to calculate the wind load transmitted to the slope. d The first mode shape of the cantilever beam representing the pine tree defined in Table 1, corresponding to the vibration frequency of 0.372. e Total force-time series at the base of the tree trunk under a Force 10 typhoon (negative values of wind force are directional values).

is capable of representing the essential mechanical response of trees under typhoons, and is mature enough that both parts (AR method and the tree swaying model) have been widely used and validated.

3.2. Contribution of trees

Trees can significantly increase slope stability owing to the root reinforcement and hydrological effects. Typhoon-associated rainstorms always have high intensity and weaken the tree stabilization effects (McGuire et al., 2016). Hence, only the root impact was considered in the stability analysis of typhoon-induced landslides. The investigation results show that pines in the study area have a vertical taproot that can extend to a depth of approximate 2 m. The main taproot with a relatively large diameter and deep growth depth can withstand axial and shear forces and interact with the soil frictionally. Here an individual root model in which each root is separated as a bar was selected based on the biological characteristics of the tree (Zhu et al., 2017):

$$c_r = \frac{T \cos(\theta) + T \sin(\theta) \tan(\varphi) + Q}{L} \quad (9)$$

where T is the tensile strength of the root and is determined from an experimental relationship: $T = 34.15d_r^{1.5829}$ (Li et al., 2010), d_r is the root diameter, θ is the angle of the root to the slip surface, Q is the shear force developed concerning the root bending strength, experienced as $\frac{T}{3}$, L is the distance between trees. The taproot is modeled as an inverted conical shape, meaning d_r decreases linearly with the embedded depth. The root diameter d_r ranged from 250 mm at the ground surface to 0 at the maximum embedded depth (2 m).

3.3. Hydrological effect

The rainfall effect can be considered by analyzing the variation of soil matric suction (shear strength) (e.g. McGuire et al., 2016; Gonzalez-Ollauri and Mickovski, 2017), which works as a component of the apparent cohesion (Ho and Fredlund, 1982). The equations are as follows:

$$\tau_f = C + (\sigma_n - u_a) \tan \varphi \quad (10)$$

$$C = c + (u_a - u_w) \tan \varphi^b \quad (11)$$

where $(\sigma_n - u_a)$ is the normal stress acting on the sliding surface, c and φ are effective shear strength parameters of the soil, $(u_a - u_w) \tan \varphi^b$ represents the additional cohesion provided by soil matric suction, and C is the total cohesion.

To predict the shear strength of unsaturated soil, many theoretical formulations have been established, and most employ a function of matric suction. Consequently, the slope stability analysis requires a soil-water characteristic curve (SWCC) or a reliable suction value of the soil (e.g. Fredlund et al., 1978; Lee et al., 2003), which is hardly available in engineering practice. Matsushi and Matsukura (2006) proposed an empirical equation obtained from a set of shear box tests using undisturbed residual soils with varying moisture contents. The test results indicated that the impact of moisture content on soil cohesion is much greater than the friction angle from the natural to saturated state, and prior studies confirmed this conclusion (Terajima et al., 2014; Chen et al., 2021). This empirical equation was used in this study to estimate shear strength reduction because of its practical advantages, especially for the rapid assessment of slope stability. The expression is as follows:

$$\tau = c_{max} \cdot e^{-\mu\omega} + \sigma \cdot \tan \varphi \quad (12)$$

where ω is the volumetric water content, φ is the friction angle, μ is the coefficient related to the susceptibility of strength reduction ($\mu > 0$), and $c_{max} \cdot e^{-\mu\omega}$ represents the soil cohesion with different water contents (c_s). Moreover, c_{max} is a hypothetical maximum value of cohesion that from a

multiple linear regression analysis, and can be determined using the measured natural cohesion (c_{sn}) and natural water content (ω_n) when μ is specific: $c_{sn} = c_{max} \cdot e^{-\mu\omega_n}$. Subsequently, the soil cohesion corresponding to different water contents was calculated and used for numerical work.

3.4. Slope stability analysis

We assessed slope stability using a modified infinite slope stability model proposed by Lu and Godt (2008), or the ratio of the downslope driving force to the resisting force holding the soil mass in place, assuming an infinite slope with a constant slope angle. The model size does not need to be defined for such an infinite slope. Moreover, trees with vertical taproots transfer the wind load to the slope through the root-soil interaction and show a similar mechanical response of a pile subjected to a lateral load. Therefore, we assume that the wind load and root enhancement only work in the region within the root-soil composite:

$$\begin{cases} F_s = \frac{c_s + (\gamma h \cos \beta + W_t) \cos \beta \tan \varphi + c_r}{(\gamma h \cos \beta + W_t) \sin \beta + F_w \cos \beta / L} & h \leq h_{root} \\ F_s = \frac{c_s + (\gamma h \cos \beta + W_t) \cos \beta \tan \varphi}{(\gamma h \cos \beta + W_t) \sin \beta} & h > h_{root} \end{cases} \quad (13)$$

The factor of safety (F_s) below 1 represents slope instability. Here, c_s is the soil cohesion, which can be determined using Eq. (12), c_r is the cohesion attributed to roots, γ is the soil unit weight, h is the depth of the sliding surface, h_{root} is the maximum depth of root-soil composite, β is the slope angle, φ is the friction angle, W_t is the weight of a tree (a homogenized value obtained by dividing the gravity of a single tree by the distance between trees L), and F_w is the wind load. In this study, the maximum value of the force-time series was selected to be F_w in Eq. (13) to simulate the maximum wind force of the gust wind, and the lowest F_s of the slope. In addition, the wind load and root reinforcement are presumably homogenized as a sort of uniform load on the slope.

A slope angle (β) of 30° and an overburden depth of 3 m was selected for the generalized model because most landslides have a slope of 20°–40° and a depth < 3 m (parts landslides have a sliding surface at the soil-rock interface). The shear strength parameters in the natural state, φ and c_{sn} , were determined from direct shear tests and published data of some recorded landslides with values of 16°–25° and 9–15 kPa, respectively (e.g. Chen et al., 2019; Yu et al., 2019; Yan et al., 2020). Additionally, Matsushi and Matsukura (2006) performed a series of shear box tests on silt-soil and provided a group of empirical parameters: volumetric water content of 0.38 with a cohesion of 13.8 kPa in the natural state, a saturated water content of 0.52, and a strength reduction coefficient value (μ) of 6.94. Investigating overburden soils showed similar results for soil component, cohesion, and volumetric water content (~ 0.33 for natural and ~ 0.54 for saturated state). Therefore, these parameters were used in this study.

To explore the influence of various factors on slope stability, we performed model simulations for several different scenarios. The relative importance of the wind load was assessed by performing model experiments using the wind load with modified wind velocities and wind directions as inputs. Force 3–14 winds on the Beaufort scale were designed for the modeling (Table 2), which account for almost all the forces in southeast coastal China, and the wind was set to blow in the downhill and uphill directions. Additionally, we isolated the impact of root enhanced cohesion on slope stability by comparing the model results with and without root reinforcements in the factor of safety calculation. Lastly, we examined the hydrological effect of soil strength reduction by modifying the soil water content within the overburden soil mass and comprehensively analyzed the slope stability combined with the wind load. Particularly, the quartile method was used to determine soil strength parameters. We performed simulations for all the scenarios using the soil shear strength corresponding to the lower

Table 2

Wind velocity used for the numerical simulation.

Wind force	3	4	5	6	7	8	9	10	11	12	13	14
Mean wind velocity (m/s)	5	7	10	13	17	20	24	28	32	36	40	45

(Q_1 , $\varphi=18.25^\circ$ and $c_{sn}=10.5$ kPa), median (Q_2 , $\varphi=20.50^\circ$ and $c_{sn}=12.0$ kPa), and upper (Q_3 , $\varphi=22.75^\circ$ and $c_{sn}=13.5$ kPa) quartile values of the given scope to ensure the universality of the generalized model (Ozeke et al., 2014).

4. Results

As trees transfer the wind load to the slope through the root-soil interaction, we assumed that the wind load-influenced region was within the root-soil composite (2 m below the ground surface). Additionally, the root reinforcement effect is only available within the zone of the root-soil composite, as trees move together with the sliding mass when the sliding surface is below the maximum depth of roots (Lin et al., 2010). Thus, a potential sliding surface was proposed at the base of the root-soil composite in view of the wind-influenced region and root reinforcement limitation. The relative importance of the wind velocity was investigated at this potential sliding surface, and the results are shown in Fig. 3.

The model-simulated F_s decreased significantly with increasing wind velocity. The results of the lower shear strength scenario (Q_1) show a F_s decrease from 1.19 (no wind) to 0.95 (Force 14 wind with a velocity of 45 m/s), and the slope becomes unstable under the typhoon with Force >12 (wind velocity >36 m/s). F_s of all the scenarios have a similar variation and decrease by 20% in a Force 14 typhoon. In particular, although the F_s of the median scenarios (Q_2) were always greater than 1, indicating stable conditions. However, it decreased to 1.08 at the extreme Force 14 typhoon, which is basically in a state of limit equilibrium, and an associated short-term rainfall may cause slope failure. Therefore, the impact of wind load cannot be ignored in the stability analysis of trees-covered slopes that are widely identified in the hilly coastal region. Typhoon-induced shallow slope failures are prone to develop near the maximum depth of the root-soil composite under extreme typhoons.

Our in-situ observations also identified many landslides developed in the windward slope, where the wind load is well known to act as a resisting force that holds the soil mass in place. Nevertheless, the driving force of the wind will still be formed in this scenario because of the inertial force generated by the tree motion (as shown in the negative parts of Fig. 2e). We extracted the minimum negative value of the wind load time series (representing the maximum driving force) as the driving force for the slope stability analysis and employed a numerical model similar to the wind speed effect (Fig. 3). F_s of all the scenarios show the

same variation trend except for a slower decreasing rate, but the slope was always in a stable condition even under the action of an extreme typhoon. Furthermore, F_s only decreased by 7% in the Force 14 typhoon, and the contribution is similar to the wind with Force 6–7 (not rare) that blows downhill. In such a scenario, wind loads can hardly fail the slope, and other factors may control the slope stability.

We further analyzed the impact of root enhanced cohesion by assessing the F_s of slopes with and without root reinforcements at 1.5–2 m below the ground surface. Lower (Q_1) shear strength and winds with Force 9–14 were employed for the research. Results indicate a much more stable condition within the reinforced zone (F_s increases by 34% at 1.5 m depth), while the root contributions rapidly decreased with the embedded depth because of the decreasing diameter (Fig. 4). Roots can provide an additional cohesion (c_r) of 0.7–7.8 kPa in the study region and is consistent with the empirical value of 2.8–6.2 kPa for pines (McGuire et al., 2016). The results also verified our assumption that the critical root-soil composite depth is a potential sliding surface for typhoon-induced landslides within the overburden soil layer.

Slopes in the study region have a two-layer structure, implying that the bedrock underlying the upper loose deposits, and the rock-soil interface becomes a relatively impermeable layer during rainfall. As the rainfall continues, the upper part of the impervious layer (rock-soil interface) gradually saturates, generating a transient saturation zone and eventually decreasing the shear strength of the failure surface. Thus, we propose another potential sliding surface at the rock-soil interface, which corresponds to our field observations. We analyzed F_s at this interface with various soil cohesions (from natural to saturated state) and the contribution of rainfall infiltration is presented in Fig. 5. F_s of slopes with different shear strength parameters rapidly decreased with the soil volumetric water content. The slope failure ($F_s < 1$) occurred in all three scenarios in a saturated state, and a F_s reduction of $\sim 29\%$ was identified when the soil volumetric water content increased from natural to saturation. The numerical results match our knowledge well, as soil cohesion is the main factor affecting the stability of shallow landslides. Eq. (12) yields estimated saturated cohesions of 3.97 kPa, 4.54 kPa, and 5.11 kPa, respectively, for the Q_1 to Q_3 scenarios, greatly reduced from the natural condition, and thus caused slope failure at the rock-soil interface.

Additional simulations were performed at the base of the root-soil composite under the combined impact of the wind load (Force 14) and rainfall infiltration (Fig. 6). The results of F_s show similar behaviors to those at the rock-soil interface as the rainfall infiltration played an

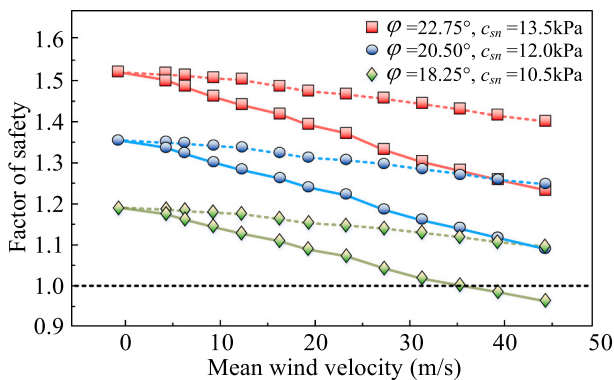


Fig. 3. Factor of safety with various wind velocities (no wind, and wind force of 3–14). Solid lines and dotted lines imply winds blow downhill and uphill, respectively.

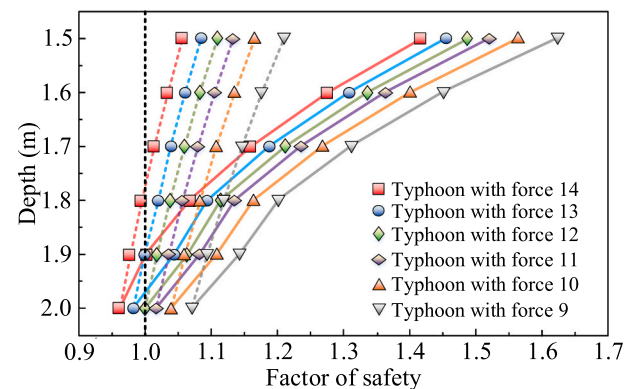


Fig. 4. Contribution of roots to slope stability. Solid lines and dotted lines imply cases with and without roots reinforcement, respectively.

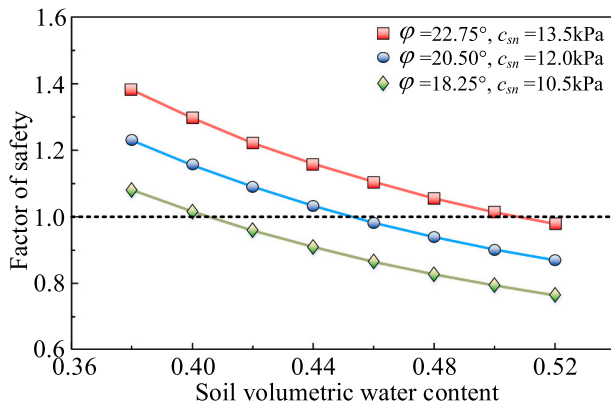


Fig. 5. Factor of safety (at the rock-soil interface) varies with soil volumetric water content, indicating the impact of rainwater infiltration.

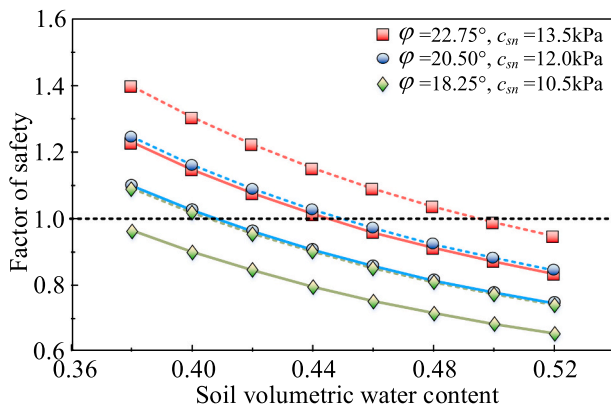


Fig. 6. Joint impact of rainfall infiltration and wind load on slope stability (at the base of root-soil composite). Solid lines and dotted lines imply winds blow downhill and uphill, respectively.

essential role in the slope stability. The model-simulated F_s of slopes reduces by 45% and 37%, respectively, for the scenarios of two wind directions with a water content from natural to saturated state, and eventually dropped to less than 1, indicating that the slope became unstable. The numerical results matched the field observations and validated our viewpoints that typhoon-induced landslides are prone to develop in the overburden soil mass, and the maximum depth of the root-soil composite and the rock-soil interface are two potential sliding surfaces.

5. Discussion

Most existing research on typhoon-induced landslides are typical case studies (especially for long-runout landslides, e.g. the Lidongcun landslide in 2016 (Ouyang et al., 2018)), or only considered the rainfall contribution to the slope stability. Hence, these landslides were usually categorized as rainfall-induced landslides, and high-intensity rainstorms were judged to be the primary cause (Chung et al., 2017; Zhang et al., 2019; Shou and Lin, 2020). This viewpoint is recognized for events that failed at the rock-soil interface, yet it is insufficient to account for landslides initiated within the overburden soil mass. More accurately, the sliding surfaces of parts shallow landslides have a position closer to the maximum depth of the root-soil composite relative to the bedrock. Notably, a violent tree-swaying movement is easily identified during a typhoon, and this phenomenon can create a driving force acting on the sliding surface. Trees covering the slope are thus vital for assessing slope stability. Our work, which focused on the impact of the trees, elucidated

this phenomenon. The driving force resulting from violent tree swaying and the root enhancement is also significantly influential for typhoon-induced landslides.

Using the numerical model, we explored how variations in wind and root enhancement influenced slope stability. Extreme typhoons can substantially increase the risk of slope failure when the wind blows downhill. Nevertheless, this effect has a relatively weak impact on slope stability with the opposite wind direction. The driving force created by the inertial force of tree swaying can hardly control the slope stability. Notably, the windthrow was neglected here as the vertical tap root could provide a great anchorage resistance, and we did not identify a wide range of windthrow near the landslide region. The numerical results further indicated the significant contribution of root enhancement, and supported a potential sliding surface near the maximum depth of the root-soil composite. The impact of roots rapidly decreased with the embedded depth because of the decreasing diameter, yet an enhanced cohesion of 0.7–7.8 kPa still remained at 1.5–2 m below the surface. The root reinforcement causes the soil below the root-soil composite to act as a weak layer and thus is prone to landslide occurrence, which is also a probable reason why parts of events occurred within the overburden soil layer. In this study, we only assessed the root contribution near the proposed potential sliding surface because the coupled action of vertical taproots and plenty of hair roots near the ground surface can significantly improve the slope stability and hardly induce a landslide.

The rock-soil interface is another potential sliding surface proposed because of the hydraulic conductivity difference between layers. Modeling slope stability at this interface suggests similar behavior to rainfall-induced landslides because the special effects of wind and root reinforcements act only within the root-soil composite. Thus, we attribute the overall F_s at the rock-soil interface to be the result of rainfall. Furthermore, the numerical results at the maximum depth of the root-soil composite also confirmed the tremendous impact of rainfall, and its combined contribution with wind load led to a higher risk of slope failures. Li et al. (2016) pointed out that the rainwater will penetrate through the upper soil mass to the rock-soil interface as a state of unsaturated infiltration, and the shallow slope is unsaturated or partly saturated. Therefore, the real water content of the upper soil during a typhoon is a certain value between the natural and saturated values. Even so, the maximum depth of the root-soil composite and rock-soil interface is reasonable to be judged as landslide-prone regions, and our field investigation verified this point.

Another interesting phenomenon is the short lag time of typhoon-induced landslides. Murielle et al. (2011) proposed the hydrological effect of preferential flow paths at the root-soil interface that resulted from biological and possible mechanical actions. This hydrological effect can greatly improve the hydraulic conductivity of the overburden soil and has been verified by many in situ dye disappearance tests or field measurements (Klos et al., 2014; Mei et al., 2018). Violent tree swaying in a typhoon (especially for trees with a large frontal area and a vertical taproot) can intensify the formation of preferential flow paths (Lin et al., 2010), which is commonly observed in post-disaster scenarios. The formatted preferential flow paths provided prone regions for rainfall infiltration, causing the saturation of the sliding surface in a relatively short time, and thus the slope could fail in short-term typhoon associated heavy rainfall. This is in contrast to the landslides induced only by slope surface infiltration because water infiltration into soil introduces an extra time lag. This particular mechanism also provided a reasonable explanation for the landslides that failed when the wind blew uphill. The wind primarily decreases the slope stability by creating preferential flow paths to increase the infiltration rate of rainwater rather than acting as a driving force. Hence, the failure mechanism of typhoon-induced landslides is highly different from landslides induced by slope surface infiltration, although rainfall plays a vital role in both scenarios. Fig. 7 shows the schematic representation of the failure mechanism of typhoon-induced landslides.

In this study, the rainfall impact was analyzed based on the cohesion

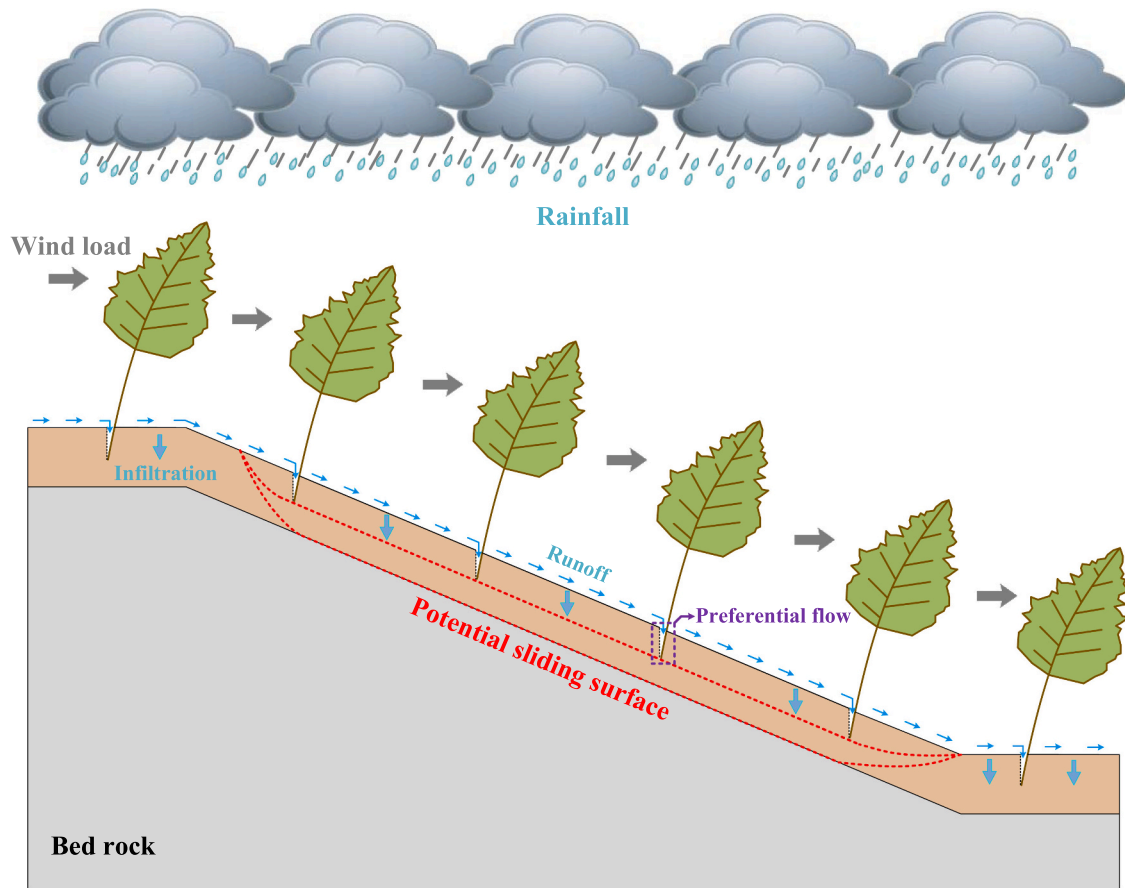


Fig. 7. Schematic representation of the failure mechanism of typhoon-induced landslides.

reduction without considering the infiltration process, especially the hydrological effect of preferential flow. This simplification cannot greatly reflect the time effect of the landslide failure process, but we acknowledge that it is difficult to establish a generalized model that covers all the factors or effects because of the complex failure mechanism of typhoon-induced landslides. Most importantly, the effects of essential factors (wind load, root reinforcement and water infiltration) on the failure mechanism were analyzed based on our proposed model. The tree-stabilizing effect on the slope is limited during a typhoon, while the combination of wind load and preferential flow will lead to a higher risk of slope failure. We further proposed two failure modes and potential sliding surfaces of typhoon-induced landslides: (1) When the wind blows downhill, the extreme typhoon may directly induce a landslide that has a potential sliding surface near the base of the root-soil composite. (2) The combined contribution of wind load and rainwater infiltration (preferential flow) can cause landslides at the rock-soil interface or near the maximum depth of the root-soil composite. We suggest that typhoon-induced landslides should be regarded as a special type of landslide in future research work and assessed based on our proposed model by combining it with field-measured data (properties of trees, soil and slope). Additionally, evaluations of landslide hazards under typhoon scenarios should account for the impact of trees, particularly concerning the wind load of typhoons and the rapid infiltration caused by preferential flow.

6. Conclusions

Typhoon-induced landslides frequently occur in coastal regions and present special development characteristics and failure mechanisms. They are mostly shallow soil landslides with a depth < 3 m and a volume $< 1000 \text{ m}^3$, and are predominantly at a trees-covered slope of 20° -

40° . Their development distribution is significantly affected by typhoons as most secondary landslides are generated within the wind circle and rainfall impacted region, and at the peak period of wind force and rainfall intensity.

We further proposed two potential sliding surfaces based on a comprehensive consideration of the impacts of wind, trees, rainfall, and slope structure: the maximum depth of root-soil composite and rock-soil interface. Numerical results of the wind load that is transmitted to the slope through the interaction with trees have an essential impact on slope stability. When the wind blows downhill, the extreme wind load may directly generate a landslide near the base of the root-soil composite (Force 14 wind causes a F_s reduction of 20%). However, this impact can hardly trigger a landslide when the wind blows uphill. Interestingly, trees only had a limited effect on stabilizing the slopes during a typhoon. The root reinforcement effect is significant in the shallow slope layer but hardly works below the root-soil composite because trees move together with the sliding mass when the root cannot penetrate the sliding surface. Additionally, the numerical results confirmed the significant impact of rainwater infiltration on slope stability. The cohesion reduction caused by rainwater significantly decreases the F_s of the slope, and its combined actions with the wind (wind load and created preferential flow path) can easily trigger a shallow landslide in a short time.

In general, typhoon-induced landslides show special features and failure mechanisms as landslides develop under the comprehensive action of wind, trees and rainfall. Thus, more studies are recommended on the contribution of wind and trees to the generation of landslides during a typhoon.

Author statement

Yu Zhuang: Conceptualization, Formal analysis, Methodology and Writing-original draft; Aiguo Xing: Conceptualization, Funding acquisition and Supervision; Yuehua Jiang, Qiang Sun and Jinkai Yan: Investigation and Resources; Yanbo Zhang: Methodology.

The authors declare that they have no known competing financial interests and agree to the submission.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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